12.10.3 Prerequisites

- PHP ≥ 8.2
- Composer
- MongoDB installed locally or MongoDB Atlas account
- Laravel 12 installed
- Basic Laravel knowledge

> *This lab uses **PHP 8.4**, **Laravel 12** and **Visual Studio Code** as the primary editor.*

12.10.4 Steps

Here are the steps to create a CRUD web application using MongoDB in Laravel 12.

12.10.4.1 Step 1: Install and Configure MongoDB

If you haven't installed MongoDB yet, you can follow the official MongoDB installation guide, https://docs.mongodb.com/manual/installation for your operating system. Alternatively, you can use a cloud service like MongoDB Atlas https://www.mongodb.com/cloud/atlas to create a free cluster.

You can also use Docker to run MongoDB. To start a MongoDB container with authentication enabled, use the following command:

```
docker run --name mongodb -d -p 27017:27017 -e MONGO_INITDB_ROOT_USERNAME=admin -e MONGO_IN
```

This command sets up MongoDB with a root user (`admin`) and password (`pass12345`). You can access MongoDB at `mongodb://admin:pass12345@localhost:27017`.

To manage MongoDB, we recommend to use MongoDB Compass, a GUI client for MongoDB. You can download it from https://www.mongodb.com/try/download/compass.